

EDITOR	Ramesh Chopra
EDITORIAL CORRESPONDENCE	Editorial Secretary Phone: 011-26810601; E-mail: editor@efyindia.com (Technical queries: efylab@efyindia.com)
SUBSCRIPTIONS & MISSING ISSUES	Customer Care Officer, EFY Phone: 011-26810601 or 02 or 03 E-mail: support@efyindia.com
BACK ISSUES, BOOKS, CDs, PCBs etc.	Kin's Spares, New Delhi Phone: 011-26371661, 26371662 E-mail: info@kinstampers.com
EXCLUSIVE NEWSSTAND DISTRIBUTOR	BH Books & Magazine Distributors Pvt Ltd, Mumbai Phone: 022-40497401, 40497402, 40497474, 40497475; Fax: 40497434 E-mail: circulation@bhworld.com
ADVERTISEMENTS NEW DELHI (HEAD OFFICE)	Sanjeev Kaushik Mobile: 0880094200 Phone: 011-26810601 or 02 or 03; Fax: 26817563 E-mail: efylab@efyindia.com
MUMBAI	McFlury's Souza Ph: 022-2495047, 24928320 Fax: 24954276; E-mail: efymumbai@efyindia.com
BENGALURU	Ms Jayashree Ph: 086-25260394, 25260023; Fax: 25260394 E-mail: efylb@efyindia.com
CHENNAI	Ms Gopali Vennari E-mail: efychennai@efyindia.com
HYDERABAD	D S Sani Mobile: 9877569691 E-mail: efylh@efyindia.com
KOLKATA	Gaurav Agarwal Mobile: 08800942001 E-mail: efylkolkata@efyindia.com
PUNE	Sandeep Shandilya Mobile: 09324299081 E-mail: efylpune@efyindia.com
GUJARAT	Sandeep Roy Mobile: 09821267855 E-mail: efylgujarat@efyindia.com
SINGAPORE	Ms Peggy Thary, Publisher Singapore Pte Ltd Phone: +65-6836 2272 Fax: +65-6297 7302 E-mail: pthary@publistatic.com, singaporemag@publistatic.com
UNITED STATES	Ms Venonique Lamareque, E&T Tech Media Phone: +1 860 536 6677 E-mail: venonique.lamareque@gmail.com
CHINA	Ms Terry Qin, Power Pioneer Group Inc. Shenzhen China, 518051 Phone: (86) 755 83729793 Fax: (86) 755 83729793 Mobile: (86) 13923802595, 18603055818 E-mail: terryqin@powerpioneer.com, qq83729793@163.com, ppg83729793@gmail.com
TAIWAN	Mr Leon Chen, J.K. Media Taipei City Phone: 886-2-87726780 ext. 10 Fax: 886-2-87726787

Printed, published and owned by Ramesh Chopra, Printed at Match Photolithographers, B-38, Okhla Industrial Area, Phase-I, New Delhi, on the 1st day of each month and published from D-671, Okhla Industrial Area, Phase-I, New Delhi 110028. Copyright 2011. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility is taken for any errors or omissions. The publisher's liability cannot be used as a defense. If a copy is found to be defective, it will be replaced free of charge. No responsibility is taken for any loss or delay in receiving the material. If it is found to be defective, it will be replaced free of charge. If any, will be replaced in a New Delhi court only.

SUBSCRIPTION RATES					
Period	News and Price (₹)	You Pay (₹)	Digital (₹)	Overseas Digital	Print
Free	3,000	2,100	—	—	—
Three	1,800	1,350	—	—	—
One	600	480	300	US\$ 10	US\$ 100

For receiving both Summer and Winter editions of EFY Directories with yellow pages (each priced at ₹ 500), please add ₹ 800 for each year (including handling and forwarding).
Kindly add ₹ 40/- for outside Delhi charges.
Please send payments only in favour of **EFY Enterprises Pvt Ltd**.
No receipt of copies may be reported to support@efyindia.com mentioning your subscription number.

CONTENTS

26 Technology Focus
Nanotechnology: The Lord of Small Things!

34 Innovation
A 'Smart' Plug That Monitors Power Consumption and Generates Alerts, Wirelessly!

Embedded
Frugal Engineering: The Mantra to Design Resource-Efficient Embedded Systems and Chips

43 Green IT
Green Computing

46 EFY Report
SMT Equipment Market in India Evolving Slowly But Steadily

56 Open Source
Building the Cloud with Open Source and Open Standards

64 Lighting
Light Emitting Diodes: Next-Generation Smart Lighting Sources

66 Automation
Barcode Systems in Use

86 Career
PCB Designing for Engineers

91 Do-It-Yourself
Simple Ways to Tap Your Laptop's Latent Talent!

94 Construction
Microcontroller-Based Light Chaser

99 Construction
Serial Communications with AVR Microcontroller

109 Software Section
JavaScript-Based ATM Simulator Interface

113 Interview
"Now we manufacture lot of things in India except high-sensitivity spectrum analyser"
— Rajesh Toshniwal, Founder & CEO of Toshniwal Enterprises Controls

115 Test & Measurement
Testing of Solar-Based Devices and Panels

Interview
"Indian market and industries will grow to become huge and energetic. Indian chemical companies and R&D centres can play as one of the world leaders" — Dr Osamu Kumagai, SVP, corporate executive, president, advanced materials laboratories, president, core device development group, Sony Corporation

Departments

WEBSITES	10
LETTERS	12
Q&As	16
TECHNOLOGY NEWS	18
CIRCUIT IDEAS	104
BOOKSHELF	122
NEW PRODUCTS	123
INDUSTRY NEWS	126
BIZ ADS	131
ELECTROONS	145
ELECTRONICS MART	146
SUBJECT & ORGANISATION INDEX	152
ADVERTISERS' INDEX	153

Style

72 First Look!

- Walkman
- Laptop
- Portable Scanner
- Digital Camera
- Speaker
- Smartphone
- Keyboard-Mouse Combo
- LED Monitor
- Mobile Phone

78 Buyers' Guide

Personal Storage Devices

NEXT ISSUE

TECHNOLOGY FOCUS

Broadband Wireless Access

BUYERS' GUIDE

Blu-Ray/HD Home Theatre Systems

SURVEY/EFY REPORT

Security Systems